A Phenomenological Statistical-Physics Framework for Distributed Consensus Phase Transitions and Thermodynamic Stability

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Moving beyond traditional cryptographic and game-theoretic analyses, we model Nakamoto consensus as a dissipative structure maintained far from thermodynamic equilibrium. By mapping macroscopic network observables onto a one-dimensional Ginzburg–Landau framework, we propose a falsifiable phenomenological description of blockchain immutability grounded in non-equilibrium statistical physics. In this description, the emergence of a canonical history appears as a continuous ordering transition, and time is treated not as an externally imposed global clock, but as an *emergent coarse-grained variable* driven by irreversible dissipation. By coupling network power, latency, and block interval into an effective temperature, the model yields a protocol-dependent upper bound on the informational volume of bytes per block ($V_{\mathrm{B,crit}}$) and anchors the ledger's macroscopic stability in the embodied exergy of the physical hardware substrate. In particular, computational validation latency (γ) imposes an irreducible constraint even in the formal limit of infinite communication bandwidth. We further interpret the Unspent Transaction Output (UTXO) set as an effective Markov blanket providing a compressed state description, and we analyze deterministic subsidy step changes (the protocol's "Halving") as non-equilibrium thermodynamic quenches. Within this framework, the observed dynamics are consistent with a persistent violation of the fluctuation–dissipation theorem, maintained by the continuous expenditure of exogenous exergy.

I. INTRODUCTION: THE THERMODYNAMICS OF DISTRIBUTED CONSENSUS

The essential concern in distributed consensus is the inability to define a canonical order for events in the absence of a global clock. Classical Byzantine Fault Tolerance (BFT) protocols typically rely on authenticated voting and logical ordering rather than an explicit thermodynamic cost; in open-membership settings, such schemes require separate Sybil-resistance assumptions [1, 2]. Physically, the double-spending problem represents a failure of time-ordering invariance: lacking an irreversible thermodynamic cost, $t \to -t$ symmetry makes history $\mathbf{h}_A = \{E_1, E_2\}$ indistinguishable from $\mathbf{h}_B = \{E_2, E_1\}$ [3]. Extracting a unique, immutable history from the stochastic noise of a peer-to-peer network requires a symmetry-breaking process driven by the energy dissipation of its internal structure [4].

Recent advances in information physics have begun mapping traditional thermodynamic ensembles onto distributed protocols, demonstrating that consensus dynamics can be modeled as irreversible processes driven by entropy reduction [5], where blockchain stability acts as a Lyapunov function [6]. Building upon this emerging consensus-physics isomorphism, we present a minimal phenomenological model of the Nakamoto protocol as a non-equilibrium thermodynamic system. To maintain an effectively objective view of the technology, our analysis intentionally decouples the treatment of financial information from the underlying physical propagation of data blocks.

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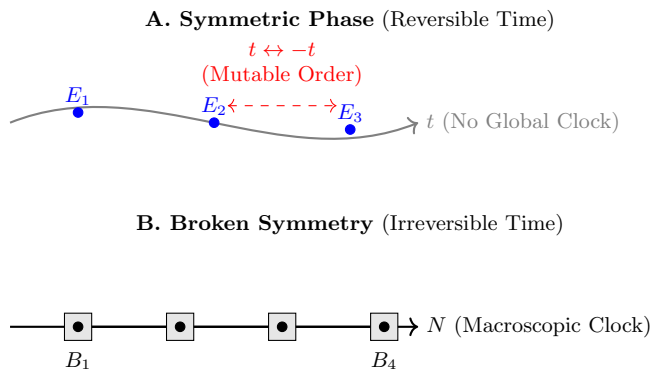

Figure 1: **Arrow of time in the effective description.** Top: without substantial dissipation, event ordering remains effectively time-symmetric and fluid. Bottom: sustained Proof-of-Work dissipation selects a stable macroscopic sequence of blocks.

Accordingly, we treat the chain tip not as a register of economic value, but as a purely informational state transition independent of transaction content.

In our effective description, the distributed ledger is conceptualized as a one-dimensional growing network $\mathcal{L}$ evolving under dissipative forcing. The development of consensus is described as a continuous phase transition between a high-entropy ("disordered") phase, the mempool, and a low-entropy ("ordered") phase, the blockchain, analogous to a liquid-crystallization process [7]. In this view, the blockchain is a dissipative structure whose low entropy is preserved far from equilibrium only through the continuous consumption of work [7, 8].

This manuscript extends a preliminary thermodynamic framework introduced in previous work [9]. While a foundational consensus-physics analogy was previously outlined, we now map alternative histories to

absorbing-state transitions via directed percolation, introduce generalized Langevin dynamics to capture heavy-tailed macroeconomic exergy shocks, establish an effective thermal bound based on computational validation latency, and present empirical observations consistent with fluctuation–dissipation violation during Halving quenches.

Before formally developing the continuous field theory and stochastic dynamics in the subsequent sections, Table I establishes the effective dictionary of isomorphisms used throughout this manuscript. It directly maps the macroscopic quantities of statistical mechanics to the observable metrics of the network.

Stat-phys quantity	**Consensus observable**	**Role in the effective description**
Local state variable $(\vec{s}_i, \theta_i)$	Local node view / branch preference	Microscopic or mesoscopic degree of freedom on the P2P network.
Global order parameter (Φ, ϕ)	Consensus coherence / normalized security density	Degree of ordering and symmetry breaking.
Effective temperature (T_{eff})	$\bar{E}_{\text{eff}}\, \tau_L/(k_B \tau_B)$	Desynchronization-driven agitation scale.
Mass gap $(m = \xi^{-1})$	Reorganization suppression scale	Inverse correlation length controlling finality.
Dissipation functional $(\Sigma(\mathbf{B}))$	Cumulative PoW expenditure	Path-weighting functional favoring highly dissipative histories.
Coupling scale $(\mathcal{E}_{\text{coupling}})$	Latency–security tension	Restoring tendency against local misalignment.
Active noise (ζ)	Block-discovery variance / exogenous shocks	Non-equilibrium stochastic forcing.
Effective Markov blanket $(\partial\mathcal{V})$	UTXO set	Compressed state description for validation.
External quench	Halving event	Deterministic protocol shock driving transient non-equilibrium response.

Table I: **Phenomenological consensus / statistical-physics dictionary.** The table summarizes the effective correspondences used in this work.

Scope, Assumptions, and Limitations

The present framework is intentionally *phenomenological*. Its purpose is not to derive Nakamoto consensus from microscopic first principles in the strict statistical-mechanical sense, but to construct a coarse-grained, dimensionally consistent description that maps observable protocol variables onto effective thermodynamic quantities.

Three scope conditions are important. First, the model targets *macroscopic* observables such as hashrate, block interval, propagation latency, and reorganization depth; it does not claim a microscopic Hamiltonian derivation of node-level decision dynamics. Second, several concepts used below—including effective temperature, Markov blanket, mass gap, and vortex unbinding—are employed as *operational analogies* or coarse-grained order-parameter descriptions rather than as literal ontological identities. Third, the analysis is specialized to Proof-of-Work systems with Nakamoto-style longest-chain selection and should not be read as a universal model of all consensus protocols.

Accordingly, the strongest claims of this manuscript are limited to statements of the following form: (i) specific dimensionless control parameters can be constructed from protocol observables; (ii) these parameters define effective ordered and disordered regimes; and (iii) the resulting framework yields falsifiable predictions concerning block-size limits, variance response to subsidy quenches, and depth-dependent suppression of reorganizations. Claims that go beyond this level—for example, exact universality-class identification or a microscopic derivation of non-equilibrium path weights—should be regarded as hypotheses for future work rather than as established results of the present paper.

II. NETWORK MODEL, STATISTICS, AND THE ARROW OF TIME

A. The Minimal Network Model and Consensus Hamiltonian

We model the blockchain as a growing one-dimensional network $\mathcal{L}$ of length $N(t)$, where each site i represents a block B_i. The macroscopic state of the system is defined by the block sequence $\mathbf{B} = \{B_0, B_1, \ldots, B_N\}$. Unlike static data structures, this network is dynamic; its geometry is determined by a stochastic growth process fighting against entropic degradation (information loss via forks).

In a classic closed thermodynamic system, the ground state emerges through the minimization of internal energy. However, the blockchain is an inherently open dissipative structure, coupled to a global exergy bath. Motivated by the Maximum Power Principle and related non-equilibrium heuristic ideas [10, 11], we model this open system as one whose long-lived histories are biased toward larger cumulative irreversible dissipation.

Consequently, to extract a unique history, we move away from equilibrium energy minimization and define a macroscopic **Dissipation Function** $\Sigma(\mathbf{B})$. In the present effective description, the ledger is biased toward trajectories associated with larger destruction of irreversible work. We formalize this extensive dissipation along a specific history path as

$$\Sigma(\mathbf{B}) = \sum_{i=0}^{N} E_{\text{eff}}(B_i) \cdot \mathbb{I}_{\text{valid}}(B_i)\,, \tag{1}$$

where $E_{\text{eff}}(B_i)$ represents the macroscopic characteristic energy dissipated to mine block B_i, and $\mathbb{I}_{\text{valid}}$ acts as a Boolean boundary constraint enforcing cryptographic validity.

We define the state space of the network Ω as the set of all possible chain permutations (forks). The probability $P(\mathbf{B})$ that a specific configuration $\mathbf{B} \in \Omega$ emerges as the macroscopic "truth" is modeled by a phenomenological

path measure weighted by cumulative dissipation:

$$P(\mathbf{B}) = \frac{1}{\mathcal{Z}_\Sigma} \exp\big[\beta_{\text{eff}}\, \Sigma(\mathbf{B})\big] \,, \tag{2}$$

with

$$\mathcal{Z}_\Sigma = \sum_{\mathbf{B}\in\Omega} \exp\big[\beta_{\text{eff}}\, \Sigma(\mathbf{B})\big] \,. \tag{3}$$

The positive sign in the exponential is intentional: unlike an equilibrium Boltzmann weight, this effective path measure assigns larger statistical weight to more dissipative histories. Because the total exergy accessible to the network is bounded at any finite time, $\Sigma(\mathbf{B})$ is bounded above for any finite history, ensuring convergence of $\mathcal{Z}_\Sigma$.

We emphasize that Eq. (3) is introduced as an *effective non-equilibrium ansatz*, not as a microscopic equilibrium ensemble derived from detailed balance. Its role is to encode, at coarse-grained level, the empirical bias of Nakamoto-style consensus toward histories with larger cumulative irreversible expenditure. In that sense, the construction is closer in spirit to a maximum-caliber or trajectory-weighting prescription than to a canonical partition function of equilibrium statistical mechanics.

B. Time-Reversal Symmetry and Sybil Degeneracy

Consider a distributed ledger system with zero energetic cost for block creation ($E_{\text{eff}} = 0$). In this regime, the system exhibits **Time-Reversal Symmetry** ($\mathcal{T}$-Symmetry). For any given history $\mathbf{B}_A$ (e.g., "E_2 precedes E_3" in Fig. 1), there exists a computationally symmetric history $\mathbf{B}_B$ (e.g., "E_3 precedes E_2") that is indistinguishable to a new observer [3]. This degeneracy implies that the "arrow of time" is undefined; the system exists in a superposition of microstates with equal statistical weight.

Mathematically, this corresponds to a system at infinite temperature ($T \to \infty$, $\beta_{\text{eff}} \to 0$). The partition function diverges, and the probability of observing a specific history $\mathbf{B}$ becomes uniform:

$$P(\mathbf{B}_A) \approx P(\mathbf{B}_B) \approx \frac{1}{|\Omega|} \,, \tag{4}$$

where $|\Omega|$ is the cardinality of the discrete state space (all possible forks). In this effective description, a "Sybil" attack corresponds to a high-degeneracy regime in which many alternative histories carry comparable statistical weight—a thermodynamic degeneracy where costless identities can generate arbitrary histories [2]. Without a symmetry-breaking field, the history entropy S_{history} is maximized, rendering the ledger purely random and devoid of information [8].

C. Information Temperature and Thermodynamic Regimes

Unlike standard equilibrium models governed by an external heat bath, the ledger's effective information temperature T_{eff} is an endogenous topological property. It effectively measures the internal stochastic agitation caused by spatial desynchronization in the P2P graph. This topological uncertainty in state propagation acts as the thermal noise resisting the alignment of the consensus field. To ensure that $\beta_{\text{eff}} = (k_B T_{\text{eff}})^{-1}$ has the dimensions of inverse energy (J^{-1}), it must compare the network's characteristic block energy scale $\bar{E}_{\text{eff}}$ with the latency-induced desynchronization timescale.

We formally define $\bar{E}_{\text{eff}}$ as the **ensemble average** of the energy required to satisfy the Proof-of-Work condition, representing the equilibrium energy scale dictated by the protocol's difficulty target. As we will formally derive in Sec. III A, this temperature isolates the tension between the work-induced crystallization time τ_B and the spatial uncertainty diffusion time (latency) τ_L. We therefore define the effective thermal energy scale by

$$k_B T_{\text{eff}} \equiv \bar{E}_{\text{eff}} \frac{\tau_L}{\tau_B} \,, \tag{5}$$

so that

$$\beta_{\text{eff}} \equiv \frac{1}{k_B T_{\text{eff}}} = \frac{1}{\bar{E}_{\text{eff}}} \frac{\tau_B}{\tau_L} \,. \tag{6}$$

By distinguishing this macroscopic reference constant $\bar{E}_{\text{eff}}$ from the instantaneous, stochastic work $E_{\text{eff}}(B_i)$ evaluated in the partition function, we preserve the thermodynamic weight of the path and prevent mathematical cancellation. Because β_{eff} possesses the dimension of J^{-1}, the product $\beta_{\text{eff}} \sum E_{\text{eff}}(B_i)$ translates into a dimensionless exponent scaled by the sequence length (block depth N). This formulation ensures that the canonical measure yields a depth-dependent potential barrier against history reorganization.

In the present phenomenological description, the critical temperature T_c is determined by the critical latency ratio Γ_c; specifically, from Eqs. (5) and (11) below, the critical temperature satisfies $k_B T_c = \bar{E}_{\text{eff}}/\Gamma_c$.

The thermodynamic phase regimes of the ledger are naturally defined as follows:

- **High Temperature ($\beta_{\text{eff}} \to 0$, $\tau_L \to \tau_B$):** Spatial diffusion entropy dominates crystallization work. The system is "liquid"; forks (thermal fluctuations) multiply, and history is mutable.
- **Low Temperature ($\beta_{\text{eff}} \to \infty$, $\tau_L \ll \tau_B$):** Negentropy flux dominates. The system "freezes" into a low-energy state, making history rigid and immutable.

In this framework, Nakamoto consensus is modeled as a driven ordering process whose coarse-grained dynamics favor low-effective-free-energy configurations,

$$\mathcal{F} = E - T_{\text{eff}} S_{\text{th}} \,, \tag{7}$$

where S_{th} represents the thermodynamic entropy, defined in analogy with standard Landau descriptions [12]. In the

quasi-stationary regime where T_{eff} varies slowly, $\mathcal{F}$ acts as an approximate Lyapunov function ($d\mathcal{F}/dt \leq 0$) for relaxation toward the dominant history [6]. Far from the adiabatic limit, this monotonicity is not guaranteed, and the system may exhibit transient free-energy increases during quench events.

In this effective description, the longest-chain rule may be interpreted as selecting histories associated with larger cumulative dissipation. Conversely, a "51% attack" is recontextualized not as a security flaw, but as a thermally induced phase transition (melting) where the localized temperature of the attack vector T_{attack} exceeds the critical binding temperature T_c of the network.

D. Spontaneous Symmetry Breaking (SSB)

The transition from the mempool to the blockchain corresponds to **Spontaneous Symmetry Breaking (SSB)**. We define the Global Order Parameter Φ as the **net magnetization** of the network, by analogy with the Ising model:

$$\Phi \equiv \left| \frac{1}{\mathcal{N}_{\text{nodes}}} \sum_{i=1}^{\mathcal{N}_{\text{nodes}}} \langle \vec{s}_i \rangle \right| , \tag{8}$$

where $\vec{s}_i$ is the "consensus spin" vector of node i.

- **Disordered Phase ($T_{\text{eff}} > T_c$):** $\Phi \approx 0$. Nodes are randomly oriented (forks). At coarse-grained level, the system may be treated as approximately rotationally symmetric in its branch-selection space, with no preferred history.
- **Ordered Phase ($T_{\text{eff}} < T_c$):** $\Phi \to 1$. Nodes spontaneously align along a single "worldline." Rotational symmetry is broken.

This phase transition is induced by the injection of low-entropy energy (electricity), which acts as an external magnetic field h_{ext} aligning the spins.

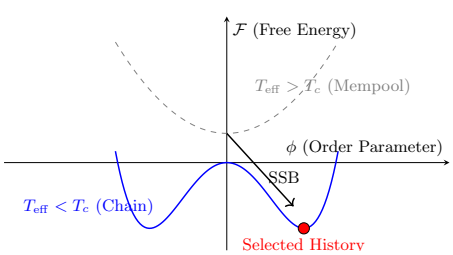

Figure 2: **Symmetry breaking in consensus.** At high effective temperature ($T_{\text{eff}} > T_c$), the free energy $\mathcal{F}$ has a single minimum at $\phi = 0$ (disorder / Sybil). Below the critical temperature ($T_{\text{eff}} < T_c$), the potential bifurcates. The system must spontaneously "roll" into one of the stable vacua (a specific history), thereby breaking the symmetry. Returning to the other vacuum (double spending) requires overcoming the potential barrier $\Delta\mathcal{F}$, which grows with confirmation depth.

E. Emergence of Thermal Time and Fluctuation Theorem

In an asynchronous distributed network, there is no absolute global clock. Time is not an *a priori* fundamental variable, but a thermodynamic emergent property [13]. We formalize this using the **Thermal Time Hypothesis** of Rovelli and Connes [14], which postulates that the flow of time is generated by the statistical state of the system itself.

For a macroscopic state described by a density matrix $\hat{\rho}$, the modular Hamiltonian is $\mathcal{H}_{\text{mod}} = -\ln \hat{\rho}$. We do not attempt a microscopic derivation of modular flow for the blockchain protocol; instead, we use the thermal-time idea heuristically and associate the advancement of internal ledger time with accumulated irreversible dissipation. In that effective sense, we use the heuristic scaling relation

$$\Delta N = c_{\text{th}} \frac{1}{k_B T_{\text{eff}}} \int_{t_1}^{t_2} \dot{Q}(t)\, dt \,, \tag{9}$$

where c_{th} is an $O(1)$ dimensionless proportionality constant whose value is fixed by the convention $\Delta N = 1$ per block.

Unlike external Newtonian time t, the internal discrete metric N advances only when irreversible heat $\dot{Q}$ is dissipated, thereby breaking detailed balance. This thermodynamic tension between continuous exergy injection and discrete topological updates dictates the emergent T_{eff} and governs macroscopic phase stability.

We quantify this time asymmetry and the resolution of the double-spending problem ($t \to -t$ symmetry) using the **Crooks Fluctuation Theorem** [15] from non-equilibrium stochastic thermodynamics. Although the Crooks relation is rigorously derived for microscopically reversible stochastic dynamics, we apply it here as a heuristic bound on the forward/reverse asymmetry of the coarse-grained ledger. The quantitative ratio should therefore be interpreted as an effective scaling rather than an exact identity.

For a ledger transition from state A to B, the ratio of the forward probability P_F (honest validation) to the reverse probability P_R (reorganization attack) is exponentially bounded by the dissipated work W exceeding the free-energy barrier $\Delta\mathcal{F}$:

$$\frac{P_F(W)}{P_R(-W)} = e^{\beta_{\text{eff}}(W - \Delta\mathcal{F})} = e^{\Delta S_{\text{irrev}}/k_B} \,, \tag{10}$$

Immutability is therefore an emergent statistical bias toward irreversible histories in this non-equilibrium description.

Throughout this manuscript, $P(\cdot)$ denotes a probability measure; the network power is always written P_{net} to avoid ambiguity.

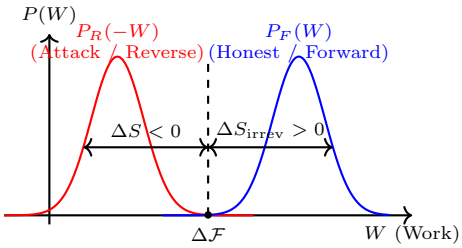

Figure 3: **Crooks fluctuation in Nakamoto consensus.** The probability distributions of honest work $P_F(W)$ and attacker reorganization work $P_R(-W)$. The intersection occurs at $\Delta\mathcal{F}$. For a deeply confirmed block, the honest network operates far to the right ($W \gg \Delta\mathcal{F}$), making the probability of a reverse trajectory exponentially vanishing.

III. THERMODYNAMIC POTENTIAL AND SYMMETRY BREAKING

We analyze ledger stability using a one-dimensional coarse-grained Landau theory in which the blockchain is represented by a scalar field $\phi(x,t)$ describing a normalized local security density along the chain direction.

A. Effective Temperature, Microscopic Latency, and the Block-Size Limit

The chain "tip" behaves as a fluid interface subject to thermal roughness (orphan blocks). We define the Effective Temperature $T_{\rm eff}$ as the energy dissipated "blindly" due to spatial desynchronization. This $T_{\rm eff}$ measures the consensus graph's stochastic agitation and must be distinguished from the substrate's true thermal temperature $T_{\rm sub}$. While $T_{\rm sub}$ dictates Landauer's limit per hash, $T_{\rm eff}$ governs macroscopic consensus stability against latency noise.

From the definition of $\beta_{\rm eff}$ [Eq. (5)], we identify a fundamental dimensionless ratio Γ comparing the thermodynamic crystallization time to the spatial entropy diffusion time:

$$\Gamma = \frac{\tau_B}{\tau_L}\,. \tag{11}$$

Since the Boltzmann-like parameter is $\beta_{\rm eff} = (k_B T_{\rm eff})^{-1}$, we obtain $\beta_{\rm eff} = \Gamma/\bar{E}_{\rm eff}$. However, network latency $\tau_L(V_B)$ is a kinetic emergent property of the P2P graph topology. Extending models by Croman *et al.* [16] and Gervais *et al.* [17], we integrate the actions required for a block of volume V_B to traverse a network of diameter $D_{\rm hop}$ (average hops to global saturation). A single-hop propagation requires geometric transmission ($\tau_{\rm link}$), bandwidth-constrained transfer, and computational validation. The single-hop time $\tau_{\rm hop}$ is

$$\tau_{\rm hop}(V_B) = \tau_{\rm link} + V_B\left(\frac{1}{C_{\rm node}} + \gamma\right), \tag{12}$$

where $\tau_{\rm link} \approx d/c$ is the geometric propagation delay, $C_{\rm node}$ is the bottleneck link capacity [Bytes/s], and γ is the validation-time coefficient [s/Byte]. Crucially, γ denotes computational friction, representing the irreducible CPU work required to verify cryptographic signatures and UTXO transitions [16].

The macroscopic network latency integrates this hop time across the graph diameter:

$$\tau_L(V_B) \approx D_{\rm hop}\,\tau_{\rm hop}(V_B) = \tau_0 + V_B D_{\rm hop}\left(\frac{1}{C_{\rm node}} + \gamma\right), \tag{13}$$

where $\tau_0 \equiv D_{\rm hop}\cdot\tau_{\rm link}$ is the incompressible geometric delay. From this, we extract the **Effective Global Channel Capacity** $C_{\rm eff}$:

$$C_{\rm eff} = \left[D_{\rm hop}\left(\frac{1}{C_{\rm node}} + \gamma\right)\right]^{-1}. \tag{14}$$

Substituting Eq. (13) into the definition of $\beta_{\rm eff}$ yields

$$\beta_{\rm eff}(V_B) = \frac{1}{\bar{E}_{\rm eff}}\left(\frac{\tau_B}{\tau_0 + V_B/C_{\rm eff}}\right). \tag{15}$$

Increasing block size is physically equivalent to heating the system. Spatial Shannon entropy increases, mechanically raising $T_{\rm eff}$. Thus, a critical block size $V_{\rm B,crit}$ exists where $T_{\rm eff}$ exceeds the phase-transition threshold ($T_{\rm eff} > T_c$). Setting $\Gamma = \Gamma_c$:

$$V_{\rm B,crit} \approx C_{\rm eff}\left(\frac{\tau_B}{\Gamma_c} - \tau_0\right). \tag{16}$$

To anchor this limit, we perform a numerical estimation. Setting a security threshold where correlation length ξ collapses (e.g., orphan rate $r_c \approx 10\%$, dictating $\Gamma_c \approx 10$), with $\tau_B = 600\,$s and $\tau_0 \approx 0.5\,$s, the critical size depends solely on $C_{\rm eff}$. Assuming a combined bottleneck yielding $C_{\rm eff} \approx 10\,$MB/s:

$$V_{\rm B,crit} \approx 10\,{\rm MB/s} \times \left(\frac{600\,{\rm s}}{10} - 0.5\,{\rm s}\right) \approx 595\,{\rm MB}\,. \tag{17}$$

Within this coarse-grained model, the asymptotic limit is controlled by the product $D_{\rm hop}\gamma$, even if communication bandwidth is taken formally large ($C_{\rm node} \to \infty$). Within the present effective model, proposals to increase block sizes to gigabytes would likely push the network beyond its thermal percolation limit. If $V_B \gg V_{\rm B,crit}$, the system abandons the ordered "crystalline" phase for a disordered "liquid" state, fracturing temporal domains.

B. The Order Parameter and Landau Free Energy

Rather than tracking discrete block topology, we analyze the system at the hydrodynamic limit via a mean-field approach. We bridge the discrete P2P network to a continuous field theory by coarse-graining highly interconnected clusters onto a spatial manifold x.

Notational Note on Time Dependence: Nakamoto consensus operates under effective macroscopic timescale

separation. Thermal relaxation ($\tau_B \approx 600\,\mathrm{s}$) is infinitesimally shorter than secular hardware evolution. We thus invoke the **adiabatic approximation**: macroscopic variables ($P_{\mathrm{net}}(t)$, $\bar{E}_{\mathrm{eff}}(t)$, $T_{\mathrm{eff}}(t)$) are quasi-stationary when analyzing the potential $\mathcal{F}$. Explicit time dependence (t) is occasionally omitted for brevity.

We use a coarse-grained scalar field $\phi(x)$ to represent the normalized local security density along the chain. We note that in one spatial dimension, the Mermin–Wagner theorem forbids spontaneous breaking of a continuous symmetry at finite temperature in equilibrium. The present description circumvents this obstruction in two ways: (i) the system is driven far from equilibrium by continuous exergy injection, and (ii) the field ϕ enters a double-well potential with discrete ($\mathbb{Z}_2$) rather than continuous degeneracy, for which the theorem does not apply.

For the present one-dimensional ledger description, the relevant Ginzburg–Landau functional is

$$\mathcal{F}[\phi] = \int dx \left[\frac{\kappa}{2}\left(\partial_x \phi\right)^2 + f(\phi)\right], \tag{18}$$

where κ is a stiffness coefficient penalizing spatial gradients of the order parameter, and

$$f(\phi) = a(T_{\mathrm{eff}})\,\phi^2 + b\,\phi^4 \tag{19}$$

is the local free-energy density. Since ϕ is dimensionless in our normalization, f, a, and b all carry the dimensions of energy per unit length in this one-dimensional description [$\mathrm{J\,m^{-1}}$], while κ carries the dimensions of energy times length [$\mathrm{J\,m}$]. In the homogeneous consensus state ($\nabla\phi \to 0$), the gradient penalty vanishes, and the system dynamics are governed entirely by this standard "Mexican hat" potential, whose coefficients are determined by network thermodynamic bounds:

1. **Instability Term and Exergetic Heat Capacity (a):** The parameter a governs symmetry breaking via T_{eff}:

$$a(T_{\mathrm{eff}}) = \chi_E (T_{\mathrm{eff}} - T_c)\,, \tag{20}$$

where $\chi_E > 0$ is an effective susceptibility converting thermal deviation into free-energy-density variation. In the present one-dimensional coarse-grained description, χ_E has units of $\mathrm{J\,m^{-1}\,K^{-1}}$.

Using $\Gamma = \tau_B/\tau_L$, the quadratic term depends on spatial latency:

$$a(\tau_L) = \mathcal{E}_{\mathrm{coupling}}\left(\Gamma_c\,\frac{\tau_L}{\tau_B} - 1\right), \tag{21}$$

where $\mathcal{E}_{\mathrm{coupling}} \equiv \chi_E T_c$ is an effective coupling free-energy density with units of $\mathrm{J\,m^{-1}}$ in this one-dimensional formulation. Substituting latency $\tau_L(V_B)$ exposes the thermodynamic coupling between validation and symmetry breaking:

$$a(V_B) = \mathcal{E}_{\mathrm{coupling}}\left\{\frac{\Gamma_c}{\tau_B}\left[\tau_0 + V_B D_{\mathrm{hop}}\left(\frac{1}{C_{\mathrm{node}}} + \gamma\right)\right] - 1\right\}. \tag{22}$$

This shows that validation friction γ enters the effective thermal control parameter directly.

- **Ordered Regime ($a < 0$):** Mining incentives overcome entropic diffusion. The trivial state $|\phi| = 0$ is unstable, and the system settles into consensus $\phi = \pm\phi_0$.
- **Disordered Regime ($a > 0$):** If friction γV_B or bandwidth limits push T_{eff} past the critical threshold, a becomes positive. The free-energy minimum shifts to the disordered state ($\phi = 0$). Exceeding this validation speed limit drives the system across the order–disorder boundary, destabilizing consensus.

2. **Saturation Term and ASIC Mass ($b > 0$):** The quartic term $b\,|\phi|^4$ provides the restoring force that prevents unbounded hashrate growth, anchoring the field theory in the material constraints of the hardware substrate.

We define **computational mass efficiency** μ_m as the ratio of average mining unit mass M_{unit} to raw hashrate H_{unit} ($\approx 1.5 \times 10^{-13}\,\mathrm{kg\cdot s}$ for 2026-era ASIC). Introducing e_{mfg} [$\mathrm{J\,kg^{-1}}$] as the specific **embodied ASIC exergy**—a composite thermodynamic constant representing the total work required to extract, refine, and assemble the active computing infrastructure—we relate the energy barrier b to this physical deployment cost via a dimensionless structural constant c_0. To ensure b properly evaluates to an energy density [$\mathrm{J\,m^{-1}}$] consistent with the 1D Ginzburg–Landau functional, we divide by the effective network spatial diameter L_{net}:

$$b = c_0\,\frac{e_{\mathrm{mfg}} \cdot \mu_m \cdot H_{\mathrm{eq}}}{L_{\mathrm{net}}}\,. \tag{23}$$

This dynamically anchors the Landau potential in the material and energetic constraints of the physical substrate [18].

The system relaxes into a stable ground state with normalized equilibrium hashrate $|\phi_0|$:

$$|\phi_0| = \sqrt{\frac{-a}{2b}}\,. \tag{24}$$

C. Phase-like Fluctuations and Consensus Rigidity

At a coarse-grained level, local disagreements about the preferred chain tip may be represented by phase-like fluctuations of an effective consensus field. In the ordered regime, such fluctuations are penalized by the stiffness term in Eq. (18), leading to an emergent rigidity against spatially extended forks. We use this language as a phenomenological description of collective synchronization on the network rather than as a claim of an exact microscopic continuous symmetry of nonce space.

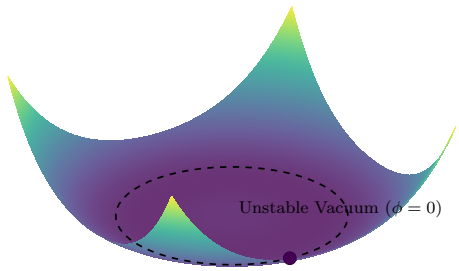

Figure 4: **The consensus potential.** The system spontaneously rolls from the unstable origin (zero security) toward the stable valley ϕ_0 (equilibrium hashrate), defined by the balance between incentive (a) and cost (b).

D. Macroscopic Energy Scale and the DAA Thermostat

To ground the macroscopic energy scale $\bar{E}_{\text{eff}}$ empirically, we define it as the *ensemble average* of the total physical work required to crystallize a single block. This statistical barrier is dictated by the Difficulty parameter $\mathcal{D}$, which imposes an expected hash count $\mathcal{N}_H(\mathcal{D})$. With $e_{\text{hash}}(t)$ representing the absolute physical energy per hash [Joules], the expected block-state energy is

$$\bar{E}_{\text{eff}}(t) = \mathcal{N}_H(\mathcal{D}) \cdot e_{\text{hash}}(t) \,. \tag{25}$$

Expressed via raw physical hashrate $H(t)$ and the target block interval τ_B:

$$\bar{E}_{\text{eff}}(t) = e_{\text{hash}}(t) \cdot H(t) \cdot \tau_B \,. \tag{26}$$

Recognizing that the product $e_{\text{hash}}(t) \cdot H(t)$ constitutes the total instantaneous network power $P_{\text{net}}(t)$ in Watts, the macroscopic scale simplifies to

$$\bar{E}_{\text{eff}}(t) = P_{\text{net}}(t) \cdot \tau_B \,. \tag{27}$$

Consequently, $\bar{E}_{\text{eff}}$ is effectively governed by the Difficulty Adjustment Algorithm (DAA), acting as a **nonlinear feedback controller**. By modulating $\mathcal{N}_H(\mathcal{D})$, the DAA scales $\bar{E}_{\text{eff}}$ to maintain the canonical $\tau_B \approx 600\,\text{s}$. In the thermodynamic limit, the DAA prevents the expected energy gap from diverging relative to total power, effectively clamping thermal fluctuations to prevent chaotic transitions.

Combining with the definition of T_{eff} [Eq. (5)] yields the operational form

$$k_B T_{\text{eff}}(t) = P_{\text{net}}(t) \cdot \tau_L \,. \tag{28}$$

The thermal noise ($k_B T_{\text{eff}}$) effectively equals the "blind" energy dissipated during the geographic latency window τ_L. This is the physical work at risk of generating an orphaned fork.

Furthermore, the relationship between block-discovery precision and energy dissipation is constrained, at least heuristically, by the **Thermodynamic Uncertainty Relation (TUR)** from modern stochastic thermodynamics [19, 20]. In its integrated hitting-time form, the TUR states that the precision of a stochastic current over an observation window requires a minimum total entropy production σ_{obs}:

$$\frac{\text{Var}(\tau_B)}{\langle \tau_B \rangle^2} \gtrsim \frac{2k_B}{\sigma_{\text{obs}}} \,, \tag{29}$$

where σ_{obs} denotes the total entropy production accumulated over the observation interval, carrying units of $\text{J}\,\text{K}^{-1}$. This suggests a thermodynamic role for the DAA. To maintain a relatively predictable block-production timescale against strong active noise, the protocol requires sustained macroscopic dissipation, consistent with thermodynamic uncertainty bounds on precision. If the absolute macroscopic hashrate $H(t)$ surged infinitely without adjustment, crystallization time would collapse ($\tau_B \to 0$), exploding the spatial entropy ratio τ_L/τ_B and melting consensus. The DAA acts as a physical thermostat, raising $\mathcal{N}_H(\mathcal{D})$ to force τ_B back to 600 s. By increasing the expected work required per block, the protocol counteracts temperature spikes in the effective description and helps preserve the ordered ledger state.

E. The Nakamoto Action and Hardware Substrate

While $\bar{E}_{\text{eff}}$ governs the macroscopic state, it rests upon a fundamental physical substrate. To construct an action-based analogy with the Principle of Least Action, we isolate the network's fundamental quantum of action: the **Nakamoto Action** h_N ($\text{J} \cdot \text{s}$). We define it kinematically at the ASIC hardware level:

$$h_N(t) = e_{\text{hash}}(t) \cdot \tau_{\text{hash}} \,, \tag{30}$$

where $e_{\text{hash}}(t)$ is absolute physical energy per hash [J] and τ_{hash} is the execution time of a single SHA-256 state transition [s]. This strips away arbitrary denominations, ensuring the product cleanly yields physical action dimensions.

A rigorous timescale separation is imperative. $h_N(t)$ operates at the *micro-canonical* substrate scale ($t \sim \tau_{\text{hash}} \sim 10^{-11}\,\text{s}$), acting as the absolute phase-space floor enforcing Landauer's limit for the logic-gate operations of a SHA-256 transition. Conversely, the ledger's *macro-canonical* dynamics operate on the scale of $\tau_B \approx 600\,\text{s}$. The macroscopic state energy $\bar{E}_{\text{eff}}$ emerges from the statistical integration of billions of micro-canonical fluctuations (the Poisson process). This coarse-graining makes consensus robust and mathematically agnostic to the specific hardware architecture, provided the system respects the physical anchor of $h_N(t)$.

Unlike Planck's constant h, $h_N(t)$ is a non-stationary, dissipative technological parameter. It dynamically revises downward with semiconductor efficiency, asymptoting toward Landauer's limit.

We define the **consensus action** over a time window

$[t_1, t_2]$ as

$$\mathcal{S}_{\mathrm{PoW}}(t_1, t_2) \equiv \tau_B \int_{t_1}^{t_2} P_{\mathrm{net}}(t)\, dt = \tau_B\, E_{\mathrm{total}}(t_1, t_2)\,, \tag{31}$$

which has the dimensions of action (J · s), since $E_{\mathrm{total}} = \int P_{\mathrm{net}}\, dt$ is the total energy dissipated over the window. The factor τ_B converts accumulated energy into an action scale by weighting each dissipated Joule by the protocol's characteristic crystallization time. In this sense, the ledger accumulates *action*, not merely data. By dissipating heat, the protocol decreases logical entropy. In this effective description, the heaviest chain corresponds to the history maximizing the accumulated quantity $\mathcal{S}_{\mathrm{PoW}}$, thereby creating a large thermodynamic barrier against reorganization.

IV. CORRELATION LENGTH AND PROBABILISTIC FINALITY

In this section, we derive the probabilistic finality of the ledger not as a geometric horizon, but as a correlation decay property in a massive field theory. We treat the blockchain as a 1D spin chain at finite temperature.

A. The Correlation Function

We define ledger stability via the two-point correlation function $G(z)$. Let $s_i^{(z)}$ be the block state at height i (confirmed vs. rejected). The probability that the state at depth z remains correlated with the current consensus tip (height N) is given by

$$G(z) = \langle s_N^{(z)} \cdot s_{N-z}^{(z)} \rangle\,. \tag{32}$$

In a massless system (zero difficulty), fluctuations would propagate infinitely, and history would be mutable at all depths ($G(z) \sim$ const). However, the Proof-of-Work mechanism introduces a "mass gap" m (the difficulty target) into the theory. In massive field theories, correlations decay exponentially with distance:

$$G(z) \sim e^{-z/\xi}\,, \tag{33}$$

where ξ is the **Correlation Length**. Because $G(z)$ measures the influence of the consensus tip on a historical block, a successful reorganization represents a total decorrelation event where the historical state $s_{N-z}^{(z)}$ is forcibly flipped by an attacker. We therefore use the decay of $G(z)$ as a proxy for probabilistic finality: as the correlation between the current tip and a block at depth z decreases, the probability of a successful reorganization is correspondingly suppressed. This length scale is determined by the ratio between the "binding energy" (honest hashrate H_H) and the "thermal noise" (attacker hashrate H_A).

B. Derivation of the Mass Gap (The "6-Block" Rule)

By mapping the correlation breakdown to Nakamoto's Gambler's Ruin formulation, the probability of a successful chain reorganization P_{reorg} can be rewritten as an effective exponential decay:

$$P_{\mathrm{reorg}}(z) = \left(\frac{q}{p}\right)^z = e^{-z \cdot \ln(p/q)}\,, \tag{34}$$

where p is the honest network hashrate share and q the attacker share ($p + q = 1$). Assuming the system operates in the ordered phase ($p > q$), we compare this to the standard exponential decay in massive field theories ($e^{-z/\xi}$) to extract the inverse correlation length (the mass gap) m:

$$m = \xi^{-1} = \ln\left(\frac{p}{q}\right). \tag{35}$$

This "mass" does not represent a fundamental gauge-boson mass, but rather an emergent dynamical inertia arising from the statistical distribution of the network's hashrate. This derivation effectively satisfies recent fluctuation bounds established for the statistical thermodynamics of reversible communicating processes, which dictate that the suppression of local entropy production in distributed algorithms manifests mathematically as an exponentially decreasing fork probability [21].

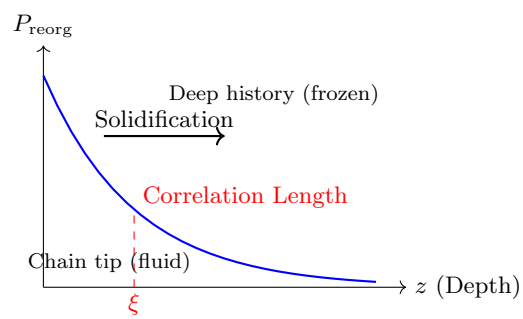

Figure 5: **Correlation decay.** The probability of reorganization decays exponentially with depth z. The system transitions from a "fluid" phase at the tip to a "frozen" solid phase in the bulk. The characteristic length scale ξ defines probabilistic finality.

Physical Interpretation:

- **Massless Phase ($m \to 0$):** As the attacker approaches parity ($p \approx q \to 0.5$), the mass gap vanishes and the correlation length diverges ($\xi \to \infty$). Finality is never reached; the system remains in a "critical state" of infinite susceptibility to reorganization.

- **Massive Phase ($m > 0$):** If $p \gg q$, the correlation length is short and history "freezes" rapidly. For a standard attacker model ($q = 0.1, p = 0.9$), the mass is $m = \ln(9) \approx 2.2$, yielding a very short

correlation length of $\xi \approx 0.45$ blocks. The canonical "6-Block Rule" ($z = 6$) thus represents a depth of $z \approx 13\xi$, suppressing the reorganization probability to an overwhelmingly secure $e^{-13} \approx 2 \times 10^{-6}$. Conversely, an attacker commanding $q = 0.38$ of the network pushes the correlation length to $\xi \approx 2$ blocks. In this regime, the 6-block depth represents only 3ξ, severely degrading the thermodynamic safety margin.

C. Nucleation of a False History

An attack on the chain is analogous to the **nucleation of a false vacuum bubble**. An attacker attempting to rewrite z blocks must create a "domain" of length z with a divergent spin configuration (an alternative history).

In standard statistical mechanics, the probability of spontaneous nucleation is exponentially suppressed by the free-energy cost $\Delta\mathcal{F}$ of the domain: $P_{\text{nucleation}} \propto e^{-\Delta\mathcal{F}/k_B T_{\text{eff}}}$. By equating this with our macroscopic reorganization probability derived above ($P_{\text{reorg}} = e^{-z/\xi}$), we can extract the exact thermodynamic potential barrier $\Delta\mathcal{F}(z)$ required to rewrite history:

$$\Delta\mathcal{F}(z) = z\left(\frac{k_B T_{\text{eff}}}{\xi}\right) = z \cdot k_B T_{\text{eff}} \ln\left(\frac{p}{q}\right). \tag{36}$$

This formulation shows that the free-energy barrier to rewrite history grows approximately linearly with depth z, while being dynamically scaled by the effective temperature of the network and the mass gap. Immutability is therefore not absolute, but **thermodynamically robust**. Rewriting history is not impossible; rather, in this effective description it becomes exponentially disfavored by irreversible dissipation, as the external work required to overcome $\Delta\mathcal{F}(z)$ quickly diverges beyond physically realizable bounds.

D. Forks as Absorbing-State Transitions (Directed Percolation)

To formalize the suppression of alternative histories, we map the blockchain growth process to a 1+1D Directed Percolation (DP) model. We define the spatial axis as the topology of network nodes and the directed temporal axis as the block height z. In this framework, a competing fork represents an "active" state propagating through the network, while the canonical longest chain acts as the **absorbing state**.

Since nodes default to the heaviest chain, abandoned forks cannot spontaneously reactivate; they are permanently absorbed into the topological vacuum. We define the control parameter for this transition as the hashrate differential $\Delta p = p - q$. With parity defining the absorbing transition, the critical percolation threshold is $p_c = 0.5$, making Δp physically identical to the proximity parameter $|p - p_c|$ used in Directed Percolation universality classes.

If the extinction of competing forks satisfies the conditions for the directed-percolation universality class—short-range interactions, a unique absorbing state, and no additional conservation laws [22]—then the survival probability $P(z)$ of a false history to depth z is expected to follow the scaling form

$$P(z) \propto z^{-\delta} \exp\left(-\frac{z}{\xi_\parallel}\right), \tag{37}$$

where $\xi_\parallel$ is the parallel correlation length, diverging near the critical point as $\xi_\parallel \propto |\Delta p|^{-\nu_\parallel}$.

When the honest network commands a majority ($p > q$), the system operates effectively in the **subcritical regime** of directed percolation. The active fork density exponentially decays toward zero. The "6-Block Rule" is thus physically equivalent to the assertion that at depth $z \approx 6$, the survival probability of the active state has been overwhelmingly attenuated by the absorbing boundary of the canonical history. If an attacker achieves $q \geq p$, the system crosses the percolation threshold into the supercritical active phase; false histories percolate infinitely through z, topological finality collapses, and the Markov blanket is irrevocably breached.

V. INFORMATION ENTROPY AND MARKOV BLANKET

The physical stability of the ledger depends on how the protocol processes and stores information against the effective limits of the communication channel, parameterized here by C_{eff}. Given the thermal percolation constraint defined previously ($V_B \ll V_{\text{B,crit}}$), the network cannot continuously propagate and verify the entirety of the historical "volume." In our effective description, the protocol mitigates this bottleneck by projecting history onto a reduced state description, which we interpret as an effective **Markov blanket**.

A. The Volume, the Markov Blanket, and RG Flow

In probabilistic modeling, a Markov blanket [23] is a boundary variable set that screens internal states from external ones. In the present context, the UTXO set plays an analogous role by providing a compressed state description sufficient for transaction validation without replaying the full history at each step. In Nakamoto consensus:

- **The Volume ($\mathcal{V}$):** The complete historical blockchain, containing all past transactions and the total dissipated thermodynamic action $\mathcal{S}_{\text{PoW}}$.

- **The Blanket ($\partial\mathcal{V}$):** The Unspent Transaction Output (UTXO) set, representing the active state of the network.

Due to finite bandwidth, the network cannot dynamically process the entire historical volume. It resolves this via a spatial and temporal Renormalization Group (RG) flow, where block validation acts as a non-invertible decimation operator $\mathcal{R}$ continuously integrating out inactive degrees of freedom:

$$\partial\mathcal{V}_t = \mathcal{R}\left(\mathcal{V}_{t-1} \cup \mathcal{B}_t\right), \tag{38}$$

where $\mathcal{B}_t$ represents the topological set of transactions within the new block. Because this many-to-one mapping is effectively irreversible, it constitutes a dissipative compression. The Shannon entropy of the volume is vastly larger than the entropy of the boundary ($S(\mathcal{V}) \gg S(\partial\mathcal{V})$). This continuous RG decimation is absolutely essential to keep the exchange volume V_B below the critical threshold $V_{\mathrm{B,crit}}$ and prevent the divergence of the system's macroscopic free energy.

B. Shannon vs. Thermodynamic Entropy

This RG projection via the Markov blanket allows us to resolve the apparent tension between Shannon information entropy [8] and Landauer thermodynamic entropy [4].

We must distinguish two coupled thermodynamic layers. First, at the *micro-canonical hardware level*, the minimum heat flux $\dot{Q}$ that must be physically dissipated into the environment to maintain the boundary against incoming information flux is effectively bounded by Landauer's limit operating at the physical substrate temperature T_{sub}:

$$\dot{Q} \geq k_B T_{\mathrm{sub}} \ln(2) \frac{dS_{\mathrm{Shannon}}}{dt}, \tag{39}$$

where T_{sub} denotes the physical substrate temperature governing Landauer erasure at the hardware level, distinct from the macroeconomic exergy-bath temperature T_{bath} introduced in Sec. VII D.

Second, to rigorously ground this constraint at the *macro-canonical consensus level*, we invoke a generalized form of the fluctuation theorem (derived from Jarzynski's equality [24] for information processing). The average irreversible macroscopic work $\langle W_{\mathrm{diss}} \rangle$ provided by the network to synchronize a new block across the topological graph must satisfy

$$\langle W_{\mathrm{diss}} \rangle \geq k_B T_{\mathrm{eff}} \ln(2)\, \Delta S_{\mathrm{Shannon}}, \tag{40}$$

where $\Delta S_{\mathrm{Shannon}}$ represents the spatial Shannon uncertainty (in bits) injected by the flow of unconfirmed new transactions. For a node to validate the network state autonomously, this massive macroscopic work—governed by the effective information temperature T_{eff}—dissipates entropy so that uncertainty regarding the UTXO boundary state is effectively brought to zero ($S_{\mathrm{Shannon}} = 0$).

C. The UTXO Set as a Physical Maxwell's Demon

The relationship between information and thermodynamic entropy, formalized by the Mass-Energy-Information (MEI) equivalence [25], becomes explicitly manifest when we model the network as a macroscopic Maxwell's Demon.

As demonstrated by Bennett [3], a Maxwell's Demon must eventually erase its memory to reset the system, paying a mandatory thermodynamic cost governed by Landauer's Principle [4]:

$$W_{\mathrm{erase}} \geq k_B T_{\mathrm{sub}} \ln(2), \tag{41}$$

per bit of erased information. The network acts as a decentralized Demon, sorting the high-entropy Mempool "gas" across the barrier. Crucially, the thermodynamic cost is not merely storing the accepted history $\mathbf{B}_A$, but the relentless evaluation and active rejection of competing histories. To reset the UTXO boundary for the next epoch, the network must continuously flush these discarded states. Rather than operating near the minimal energetic bounds suggested by Landauer's principle, the protocol utilizes the DAA to artificially inflate this required dissipation by trillions of orders of magnitude. This deliberately massive macroscopic heat exhaust ($\dot{Q}_{\mathrm{diss}}$) ensures that the accumulated thermodynamic "escape velocity" (the integrated $\Delta\mathcal{F}$ barrier) required for an attacker to rewrite history remains physically unattainable. To reduce uncertainty and synchronize a canonical block across the network, the protocol dissipates this energy:

$$\dot{Q}_{\mathrm{diss}} \sim \frac{\bar{E}_{\mathrm{eff}}}{\tau_B}. \tag{42}$$

In practice this dissipation is vastly larger than the minimal Landauer scale,

$$\dot{Q}_{\mathrm{diss}} \gg k_B T_{\mathrm{sub}} \ln(2)\, \frac{dS_{\mathrm{Shannon}}}{dt}. \tag{43}$$

This formulation provides a physical motivation for the phenomenological inertia scale introduced in the conclusion. The colossal energy consumed by the network is not "wasted" computation; it is the fundamental thermodynamic work required to perform macroscopic uncertainty erasure. As recently argued through the lens of media archaeology, this mechanism redefines computational work effectively as "heat-work"—an irrecoverable, exhaustible thermodynamic asset that physically secures the data structure by preventing costless reversibility [18]. Immutability is physically sustained because reversing the ledger would require the spontaneous, localized absorption of this exact macroscopic heat back from the environment—a statistical impossibility.

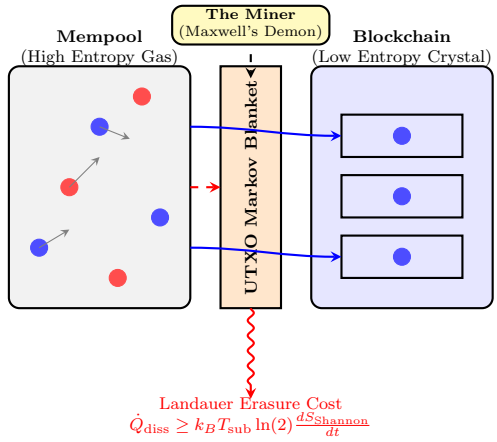

Figure 6: **Nakamoto consensus as a macroscopic Maxwell's Demon.** The network acts as a thermodynamic Demon, measuring the Mempool (high Shannon entropy) and sorting valid transactions across the UTXO boundary (the Markov Blanket). To reset its memory for the next epoch and enforce a singular history, the Demon must perform the work of *erasure* on all rejected alternative states. The protocol pays this mandatory Landauer cost via Proof of Work, dissipating massive heat $\dot{Q}_{\rm diss}$ into the environment to sustain informational certainty.

D. Blanket Saturation and Channel Limit

The security of this Markov blanket is intrinsically linked to the equations of information fluid dynamics. State updates $\partial\mathcal{V}$ (new blocks V_B) must pass through the effective communication channel $C_{\rm eff}$. If the boundary update rate approaches channel capacity ($V_B \to V_{\rm B,crit}$), the Markov property collapses globally. Nodes can no longer synchronize the UTXO surface within the allotted time τ_B. Shannon uncertainty explodes locally, effective temperature $T_{\rm eff}$ crosses the critical threshold, and consensus fractures into multiple isolated domains. History rigidity is therefore maintained only when the boundary update size remains below the spatial dissipation capacity of the network.

VI. TOPOLOGICAL STABILITY OF THE NETWORK GRAPH

While the 1D Directed Percolation model isolates longitudinal stability (temporal depth), robustness against partition attacks requires analyzing transverse stability—the long-range phase order across the spatial peer-to-peer network. We model this using the **XY Model** on a **"small-world"** graph [26].

This architecture is characterized by two fundamental physical and mathematical properties:

1. **A high clustering coefficient:** Nodes form highly interconnected local clusters, ensuring massive information redundancy at the regional level.

2. **A very short average distance (shortcuts):** A few long-distance random connections link distant clusters. On average, a transaction or new block can reach any node on the globe in just a few hops.

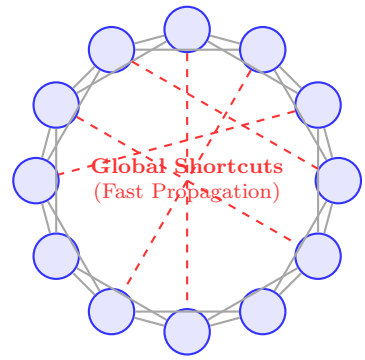

Figure 7: The "Small-World" topology of a distributed consensus network. Local links (gray) ensure redundancy and fault tolerance, while long-distance connections (dashed red) allow information to traverse the network in few hops, minimizing propagation delay.

A. The Consensus Phase Field

We assign a continuous phase variable $\theta_i \in [0, 2\pi)$ to each node i, representing its local view of the ledger tip. Interaction energy between peers seeks to minimize phase differences (consensus):

$$\mathcal{H}_{XY} = -\bar{J} \sum_{\langle i,j \rangle \in \mathcal{M}} \cos(\theta_i - \theta_j)\,, \tag{44}$$

where $\mathcal{M}$ represents the specific subgraph of active mining nodes. Crucially, non-mining nodes (which merely relay data) do not contribute to the topological rigidity of the consensus; they act as passive observers adopting the local phase field.

To ensure internal consistency of the model, we define the mean coupling constant $\bar{J}$ not as an arbitrary free parameter, but as the average local action density injected into the active network:

$$\bar{J} \approx h_N(t) \cdot \langle H_i \rangle_{\mathcal{M}}\,, \tag{45}$$

where $\langle H_i \rangle_{\mathcal{M}}$ represents the mean absolute hashrate (in s^{-1}) of an active mining cluster. This formulation explicitly bridges the micro-canonical and macro-canonical scales: the microscopic switching time of a silicon logic gate ($\tau_{\rm hash}$ embedded within h_N) serves as the fundamental quantum of action that physically finances the macroscopic phase alignment (θ) of decentralized nodes across the globe. It directly links the topological rigidity of consensus (vortex resistance) to actual hardware-level energy dissipation.

- **Ground State ($\nabla\theta = 0$):** Global Consensus. All nodes agree on the tip.

- **Excited State ($\nabla\theta \neq 0$):** A Fork. The network is split into domains with different phases.

B. Forks as Topological Defects (Vortices)

In two-dimensional systems (like the P2P overlay network), thermal fluctuations can generate topological defects known as **vortices**. Similar to localized phase alignment observed in the Axelrod model of cultural convergence [27], a vortex corresponds to a closed loop of nodes maintaining divergent views on the chain state. Furthermore, the thermodynamic suppression of these topological defects directly parallels the phase-synchronization boundaries observed in recent mean-field models of large-scale epidemic consensus [28]:

$$\oint_{\mathcal{C}} \nabla\theta \cdot d\mathbf{l} = 2\pi n, \quad n \in \mathbb{Z}\,. \tag{46}$$

If $n \neq 0$, the loop encloses a singularity (a persistent fork). Because the energetic cost of such a defect depends on the effective dimensionality of the network, isolated forks are suppressed differently depending on the structural health of the P2P graph.

C. Effective Dimensionality and the Kosterlitz–Thouless Bound

As a heuristic description of spatial consensus loss under severe network partitioning, we analyze vortex dynamics using the **Kosterlitz–Thouless (KT)** formalism. Before applying 2D topological physics to a scale-free graph, it is necessary to resolve an apparent topological paradox. The Mermin–Wagner Theorem [29] states that spontaneous breaking of a continuous symmetry is impossible at finite temperature for *equilibrium* systems of dimension $d \leq 2$. While the network operates as a Non-Equilibrium Steady State (NESS) driven by active noise—a condition known to allow behavior that can circumvent standard Mermin–Wagner expectations in certain active-matter systems—the primary topological defense mechanism lies in the dynamic effective dimensionality d_{eff} of the graph, extracted from the scaling law linking the number of accessible nodes $\mathcal{N}_{\text{nodes}}(r)$ within a topological radius r (number of hops):

$$\mathcal{N}_{\text{nodes}}(r) \propto r^{d_{\text{eff}}}\,. \tag{47}$$

In a healthy state, the P2P graph's effective dimensionality approaches infinity ($d_{\text{eff}} \to \infty$) due to global intercontinental shortcuts. However, when a severe network partition occurs (e.g., the isolation of a continental grid), the routing topology severely degrades. Stripped of its global shortcuts, the propagation layer is forced to embed much more closely onto the physical 2D surface of the Earth. Empirical studies of scale-free routing topologies reveal that under massive link failure, the spectral dimension collapses toward $d_s \approx 2$ [30]. Accordingly, a severely fragmented network may be approximated, as an idealized limiting case, by an effectively two-dimensional propagation layer.

In this critically reduced 2D regime, the energy of an isolated vortex (a partitioned fork) grows logarithmically with network size L:

$$E_{\text{vortex}} \approx \pi \bar{J} \ln\left(\frac{L}{a_0}\right), \tag{48}$$

where a_0 is the lattice constant (the minimum discrete distance, typically one hop). Simultaneously, the vortex core can be located on any of the $(L/a_0)^2$ sites of the two-dimensional lattice. The system thus acquires a configurational entropy:

$$S_{\text{vortex}} \approx k_B \ln\left[\left(\frac{L}{a_0}\right)^2\right] = 2k_B \ln\left(\frac{L}{a_0}\right). \tag{49}$$

Consensus stability against free vortex formation is governed by the variation in free energy $\Delta\mathcal{F} = E_{\text{vortex}} - T_{\text{eff}} S_{\text{vortex}}$:

$$\Delta\mathcal{F} = 2k_B(T_{\text{KT}} - T_{\text{eff}}) \ln\left(\frac{L}{a_0}\right). \tag{50}$$

This balance equation immediately reveals the existence of an effective critical partition temperature $T_{\text{KT}} = \pi\bar{J}/(2k_B)$, representing the topological unbinding threshold where global phase coherence is lost. Survival of the canonical ledger truth then boils down to a topological phase transition:

- **Ordered Phase ($T_{\text{eff}} < T_{\text{KT}}$):** $\Delta\mathcal{F} > 0$. Creating a free vortex requires infinite energy in the thermodynamic limit ($L \to \infty$). Phase fluctuations are confined; vortices and anti-vortices appear in bound pairs of finite energy. Long-range order and global consensus are maintained.
- **Partition Phase ($T_{\text{eff}} > T_{\text{KT}}$):** $\Delta\mathcal{F} < 0$. Configurational entropy outweighs binding energy $\bar{J}$. The proliferation of isolated vortices becomes thermodynamically favorable and spontaneous. Uncertainty over history curls in on itself (Fig. 8), destroying global phase coherence and irreparably fracturing the ledger.

The "Small-World" topology is therefore the indispensable structural mechanism that perpetually maintains the network in the $T_{\text{eff}} \ll T_{\text{KT}}$ regime, thereby dodging the topological collapse dictated by two-dimensional physics.

VII. NON-EQUILIBRIUM DYNAMICS: EXERGETIC EXPANSION AND THE "HALVING" QUENCH

While Sec. VI addresses spatial stability, the network must also survive severe temporal shocks. Because the Nakamoto protocol operates as an open, non-equilibrium system, it is fundamentally coupled to the global energy

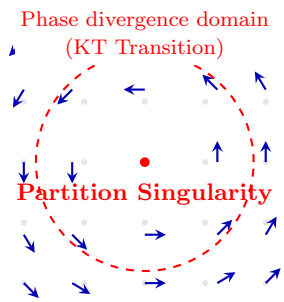

Figure 8: **Topological defect in the P2P network.** Each arrow represents the "consensus spin" of a node. A free vortex ($n = 1$) breaks long-range order, rendering global consensus physically impossible.

infrastructure. It continuously hunts **exergy** (available, unvalued energy, such as stranded energy or gas flaring) to sustain its macroscopic order.

The "Halving" event is therefore not a simple parametric update leading to a lower energy equilibrium; it is a thermodynamic shock applied to an expanding system. We model this event as a **Thermodynamic Quench** superimposed on a non-stationary potential.

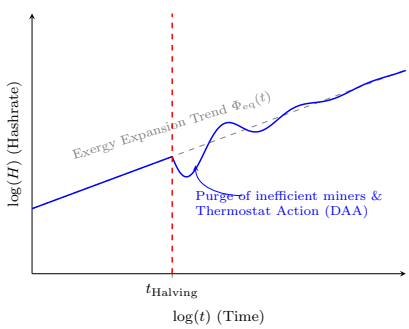

Figure 9: **Quench on non-stationary background.** Order parameter Φ is sustained by secular growth (exergy hunting). The Halving imposes an abrupt perturbation. Miner inertia and thermostat intervention (DAA) create a transient plateau or dip (damped relaxation) before the growth trend resumes.

A. The Exergy-Driven Potential

The homogeneous effective potential $\mathcal{U}(\Phi)$ is driven by the total **exergetic chemical potential** $\mu(t)$, defined as the macroscopic pool of available thermodynamic work (e.g., stranded energy in Joules). This potential features a secular growth component decomposed as

$$\mu(t) = \mu_{\text{subsidy}}(t) + \mu_{\text{exergy}}(t)\,. \tag{51}$$

The subsidy acts as a cascaded thermodynamic quench, formalized via a discrete floor function: $\mu_{\text{subsidy}}(t) = \mu_{\text{genesis}} \cdot 2^{-\lfloor t/\tau_H \rfloor}$. For a single Halving at t_H, this reduces to a Heaviside step function where the economic reward acts as a proxy for accessible Joules.

To respect planetary boundaries, $\mu_{\text{exergy}}(t)$ is ultimately constrained by the biosphere's maximum accessible exergy $\mu_{\max}$, obeying a modified logistic equation:

$$\frac{d\mu_{\text{exergy}}}{dt} = r_{\text{tech}}(t) \cdot \mu_{\text{exergy}} \left(1 - \frac{\mu_{\text{exergy}}}{\mu_{\max}}\right). \tag{52}$$

Empirically, the innovation rate $r_{\text{tech}}(t)$ exhibits diminishing returns ($\propto \nu_H/t$), averting exponential explosion. While hashrate grows super-linearly, actual power $P_{\text{net}}(t)$ remains effectively bounded by the asymptotic decay of $e_{\text{hash}}(t)$ toward Landauer's limit.

Empirically validating this expansion, we analyze hashrate evolution excluding the pre-2010 transient phase. Operating far from planetary saturation ($\mu_{\text{exergy}} \ll \mu_{\max}$), the time-dependent innovation rate forces the network into an effective temporal power-law attractor (Fig. 10):

$$H(t) = A \cdot \left(\frac{t}{t_0}\right)^{\nu_H}, \tag{53}$$

where t is the time since genesis and $t_0 = 1\,\text{day}$ is the characteristic reference time. Regression in log–log space yields an effective scaling exponent $\nu_H = 11.59 \pm 0.31$ and an amplitude $A = (6.7 \pm 3.1) \times 10^{-35}\,\text{TH/s}$ (95% confidence intervals from ordinary least squares on the log-transformed data). We report this fit as a descriptive secular trend rather than as evidence of an exact asymptotic law; parameter uncertainty and model comparison against alternative growth forms (exponential, stretched exponential) remain important directions for future work.

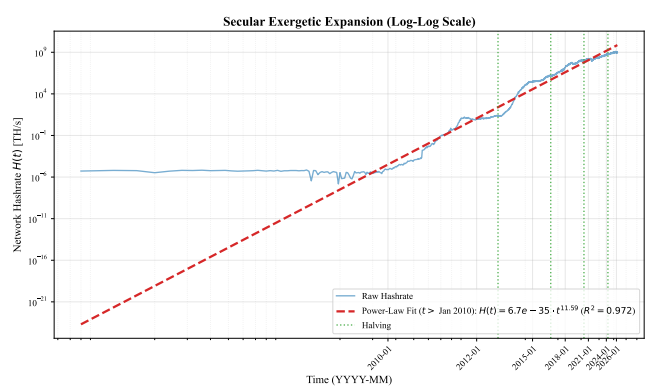

Figure 10: **Secular exergetic expansion.** The network's hashrate $H(t)$ aligns with a power-law attractor $H(t) \propto t^{\nu_H}$ with $\nu_H \approx 11.59$ ($R^2 = 0.972$). This expansion is consistent with the qualitative expectations of the Maximum Power Principle, insofar as the network continues to expand exergy capture despite periodic Halving quenches.

B. Non-Equilibrium Stochastic Dynamics and FDT Violation

Assuming the network operates deep within the ordered phase ($T_{\text{eff}} \ll T_{\text{KT}}$), the phase-like mode is taken

to remain effectively pinned ($\nabla\theta \approx 0$), allowing us to reduce the spatial field to a real global temporal order parameter,

$$\Phi(t) \equiv \langle \phi(x,t) \rangle_x \,. \tag{54}$$

By treating $\Phi(t)$ as the normalized deviation ($H/H_{\rm eq}$) around the moving adiabatic equilibrium, the spatially integrated homogeneous potential $\mathcal{U}(\Phi)$ becomes quasi-stationary during the short temporal window of a quench. Trapped in the positive vacuum ($+\phi_0$), $\Phi(t)$ is effectively confined to $\mathbb{R}^+$. Because the system operates deep within this well ($\Phi \approx \Phi_0 \gg 0$), the probability of zero-crossing is exponentially suppressed. This justifies using standard additive noise with a formal reflective boundary at $\Phi = 0$ to prevent unphysical negative hashrate excursions.

To capture the network's pre-heating phase driven by forward-looking capital expenditure, we inject a phenomenological anticipatory driving force $\mathcal{A}(t)$ into a generalized Langevin equation. This term encodes observed miner behavior—hardware pre-deployment in anticipation of subsidy changes—and is not derived from the thermodynamic framework itself:

$$\frac{d\Phi(t)}{dt} = -\Lambda_0 \frac{\partial \mathcal{U}(\Phi)}{\partial \Phi} + \mathcal{A}(t) + \zeta(t) \,, \tag{55}$$

where

$$\mathcal{U}(\Phi) = \alpha_{\rm h}(T_{\rm eff})\, \Phi^2 + b_{\rm h}\, \Phi^4 \tag{56}$$

is a homogeneous effective potential for the global order parameter. Here $\alpha_{\rm h}$ and $b_{\rm h}$ carry dimensions of absolute energy [J], representing the spatial integration of the local free-energy density over the network diameter ($\alpha_{\rm h} = a \cdot L_{\rm net}$ and $b_{\rm h} = b \cdot L_{\rm net}$), bridging the continuous 1D field theory to a lumped-parameter global dynamic.

The anticipatory force $\mathcal{A}(t)$ evaluates the expected gradient of the exergy potential over a planning horizon $\tau_{\rm plan}$:

$$\mathcal{A}(t) = -g_{\rm ant} \int_t^{t+\tau_{\rm plan}} e^{-\lambda(t'-t)} \mathbb{E}\left[\frac{\partial \mu_{\rm subsidy}(t')}{\partial t'} \right] dt' \,. \tag{57}$$

Because the Halving subsidy collapse is a deterministic downward step function, its temporal derivative constitutes a negative Dirac delta function $-\Delta\mu\, \delta(t'-t_H)$. The negative sign in our driving-force formulation correctly inverts this anticipated loss, meaning the integral mathematically evaluates to a positive, macroscopic anticipatory pulse $\mathcal{A}(t) \propto \Delta\mu\, e^{-\lambda(t_H - t)}$ leading up to t_H. This physically manifests as an exergy run-up that accelerates block discovery as miners rush to extract value before the quench.

Hunting exergy far from equilibrium, the network violates the Fluctuation–Dissipation Theorem (FDT) ($\dot{S}_i > 0$). To formalize this Non-Equilibrium Steady State (NESS), we quantify the deviation using the correlation function $C_\Phi(t,t') = \langle \delta\Phi(t) \delta\Phi(t') \rangle$ and the linear response function $R(t,t')$ to a localized exergy injection $h(t')$. This yields the Fluctuation–Dissipation Ratio (FDR) $X(t,t')$:

$$R(t,t') = -\frac{X(t,t')}{k_B T_{\rm bath}} \frac{\partial C_\Phi(t,t')}{\partial t}\, \Theta(t-t') \,. \tag{58}$$

The **macroeconomic exergy-bath temperature** $T_{\rm bath}$ is an effective temperature-like scale capturing stochastic fluctuations in stranded-energy availability. We define this thermal energy proportionally to the standard deviation of the exergetic chemical potential $\mu(t)$ over a macroscopic window $\tau_{\rm macro}$:

$$k_B T_{\rm bath} \equiv c_m \sqrt{\langle (\mu(t) - \langle \mu \rangle)^2 \rangle_{\tau_{\rm macro}}} \,. \tag{59}$$

As miners operate at margins dictated by this stochastic availability, the temporal variance of $\mu(t)$ acts as the kinetic driver. Dominated by **active noise** $\zeta(t)$ from macroeconomic shocks, time-translation invariance breaks ($X < 1$). We define the **effective activity temperature** $T_{\rm act} \equiv T_{\rm bath}/X$ and introduce the corresponding active diffusion scale

$$D_{\rm act} \equiv \Lambda_0 k_B \left(T_{\rm act} - T_{\rm bath} \right) , \tag{60}$$

which has units of $\rm s^{-1}$, consistent with the noise correlation $\langle \zeta(t)\zeta(t') \rangle$. The noise correlation can then be written as

$$\langle \zeta(t)\zeta(t') \rangle = 2\Lambda_0 k_B T_{\rm act}\, \delta(t-t') \,, \tag{61}$$

so that

$$X = \left(1 + \frac{D_{\rm act}}{k_B \Lambda_0 T_{\rm bath}} \right)^{-1} . \tag{62}$$

The network's elevated temperature directly reflects the mechanical work expended to maintain the UTXO Markov blanket.

For extreme non-Gaussian jumps, active noise generalizes to heavy-tailed Lévy ν-stable noise $\zeta_\nu(t)$, requiring a power-law memory kernel. However, our empirical metrology in Sec. VII C isolates discrete phase windows where variance remains functionally finite, permitting standard statistical validation of the anticipatory and quench dynamics.

C. Thermodynamic Halving Quench and Thermostat (DAA)

The Halving imposes an **abrupt phase transition** on this tensioned system. The subsidy collapse renders the state $\Phi(t_H)$ metastable for inefficient operators. Due to sunk CapEx (thermodynamic inertia), miners unplug only when OpEx exceeds profitability. The resulting hashrate drop against a rigid Difficulty $\mathcal{D}$ diverges mean block wait time $\langle \Delta t \rangle$. The system escapes this transient entropic spiral via the Difficulty Adjustment Algorithm (DAA). Acting as a non-equilibrium thermostat [31], the DAA undergoes damped oscillations before restoring the system's mass toward a new metastable equilibrium.

Validation: Sensitivity Analysis of Anticipation and Quench

Analyzing the inter-block wait time Δt for the Nakamoto protocol's four historical Halvings provides empirical validation of the anticipatory force $\mathcal{A}(t)$. To prevent arbitrary phase boundaries, we performed a statistical sensitivity analysis, sweeping the anticipation lookback horizon k from 1 to 14 epochs (up to $\sim$6.5 months prior to the Halving) to isolate the maximal statistical divergence via Fisher and Levene tests.

As demonstrated in Fig. 11 and Table II, the optimized phase integration reveals a profound evolution in network thermodynamics corresponding to industrial maturity:

1. **Industrial Compression of Anticipation:** In early cycles (2012, 2016), the optimal anticipation horizon extended to 14 epochs, illustrating a slow, secular hardware integration process. Conversely, in mature markets (2020, 2024), anticipation compressed into acute pre-heating windows (3 epochs and 1 epoch, respectively). This is consistent with the interpretation that mature, industrialized supply chains can concentrate CapEx closer to the quench date.

2. **Pure Quench Divergence (2020):** Following a 3-epoch anticipation phase, the 2020 event triggered a +54.71% increase in active-noise variance $\mathrm{Var}(\Delta t)$ ($p = 1.27 \times 10^{-21}$).

3. **Frustrated Divergence (2024):** The network exhibited a highly compressed, 1-epoch anticipation phase where variance dropped to $87.33\,\mathrm{s}^2$. The post-Halving quench pushed variance up by +11.28%. However, the rapid DAA intervention (672 blocks) truncated the relaxation phase, rendering the variance spike statistically marginal ($p = 0.085$) and directly illustrating the damping power of the protocol's feedback loop.

This multi-halving analysis suggests that the ledger responds to shocks not merely through production drops, but through a non-equilibrium regime shaped by both forward-looking exergy injection and rapid thermostat intervention.

Halving Year	Opt. Horizon (k Epochs)	Quench Blocks	Anticipation Var(Δt)	Quench Var(Δt)	Statistical Tests %Δ Var	Fisher (p)
2012	14	1680	99.55	125.29	+25.86%	< 0.001
2016	14	1344	90.65	100.63	+11.00%	0.0069
2020	3	1008	90.21	139.56	+54.71%	< 0.0001
2024	1	672	87.33	97.18	+11.28%	0.0853

Table II: **Thermodynamic quench metrology (sensitivity analysis).** Comparison of block-time variance between the optimized Anticipation Phase (representing force $\mathcal{A}(t)$) and the transient Quench Phase. The shift from long horizons ($k = 14$) to compressed horizons ($k = 1, 3$) demonstrates the maturation of the hardware deployment lifecycle. Variance values are reported in units of s^2.

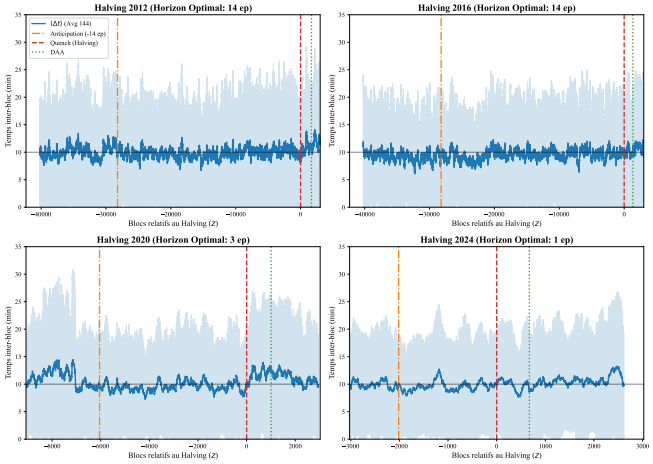

Figure 11: **Empirical relaxation dynamics of the thermodynamic quench.** Moving average of inter-block time $\langle \Delta t \rangle$ (solid blue) and thermal noise variance $\pm 1\sigma$ (shaded) relative to the Halving (dashed red). The phase boundaries reflect the optimal anticipation horizon isolated by our statistical sensitivity analysis.

D. Testable Predictions and Operational Diagnostics

Because the present framework is phenomenological, its value depends on whether it produces operationally testable signatures. The following diagnostics summarize the principal predictions of the model:

1. **Latency-controlled effective temperature.** The definition $k_B T_{\mathrm{eff}} = \bar{E}_{\mathrm{eff}}\, \tau_L/\tau_B$ [Eq. (5)] predicts that, at fixed $\bar{E}_{\mathrm{eff}}$ and τ_B, increases in propagation latency τ_L should systematically correlate with elevated orphan risk and reduced local ordering stability.

2. **Protocol-dependent upper bound on block volume.** The bound $V_{\mathrm{B,crit}} \approx C_{\mathrm{eff}}(\tau_B/\Gamma_c - \tau_0)$ [Eq. (16)] predicts that any protocol modification that increases validation friction γ or effective hop diameter D_{hop} should reduce the admissible block volume before large-scale desynchronization appears.

3. **Depth-dependent reorganization suppression.** In the ordered regime, the model predicts $P_{\mathrm{reorg}}(z) \sim e^{-z/\xi}$ with $\xi^{-1} = \ln(p/q)$ [Eq. (35)], implying that empirical estimates of reorganization risk should collapse onto an exponential depth law when conditioned on the honest/attacker hashrate ratio.

4. **Quench-induced excess variance.** Deterministic subsidy reductions should generate transient increases in block-time variance, with amplitude modulated by the anticipation horizon and by the damping action of the DAA.

5. **Compression of anticipation horizons with industrial maturation.** If the anticipatory force

$\mathcal{A}(t)$ captures real forward-looking deployment dynamics, then the optimal pre-quench horizon extracted from historical data should shorten as mining supply chains become more industrialized and capital deployment becomes more responsive.

These predictions are deliberately stated in operational form so that they may be confronted with simulation studies, historical chain data, or controlled stress tests on experimental Nakamoto-style networks.

VIII. CONCLUSION: THE PHYSICS OF DISTRIBUTED CONSENSUS

We have developed a minimal physical model in which the Nakamoto blockchain is treated as a **dissipative structure** maintained far from equilibrium. Mapping the ledger to a one-dimensional network reveals consensus as a **continuous phase transition**, where the hashrate order parameter spontaneously breaks local time-inversion symmetry.

A. Model Summary

Our analysis yields five key physical perspectives:

1. **Thermal Time and Information Temperature:** Time within the consensus is modeled not as an externally imposed global variable, but as an emergent coarse-grained quantity associated with irreversible dissipation. This formalizes the effective network temperature (T_{eff}), directly bridging information physics with macroscopic thermodynamic stability.

2. **Thermodynamic Depth:** Ledger "immutability" is not absolute but probabilistic, governed by a correlation length ξ that decays exponentially with cumulative work. This resolves the "double spending" problem as the suppression of thermal fluctuations.

3. **Crystallization and Spatial Limit:** Mining is a phase boundary transitioning information from a "liquid" to a "crystalline" state. This stability is conditioned by a dimensionless ratio Γ that imposes an effective limit on block size to avoid thermal percolation.

4. **Dynamic Stability:** Network resilience to macroeconomic shocks ("Halving") is explained by first-order quench dynamics, effectively damped by a protocol thermostat (DAA) that provides a restorative force toward the new equilibrium.

5. **Maxwell's Demon and Informational Erasure:** The network functions as a decentralized Maxwell's Demon, sorting the mempool's high-entropy "gas" into the UTXO set's low-entropy "crystal." To minimize Shannon uncertainty at this Markov blanket, the protocol must maximize thermodynamic entropy production by dissipating the mandatory Landauer cost of state erasure.

B. Thermodynamic Inertia and a Phenomenological Mass Analogy

Our framework suggests a phenomenological notion of ledger inertia at macroscopic scale, analogous to a mass-like resistance to history reversal. While assigning relativistic inertia to abstract information remains speculative, the present model motivates an effective inertia generated by cumulative dissipation under Proof of Work.

We define the phenomenological history inertia scale $M_{\text{eff}}^{(\text{phen})}$ as a thermodynamic resistance to temporal reversibility, derived from total dissipated work. Since macroscopic network power is $P_{\text{net}}(t) = \bar{E}_{\text{eff}}(t)/\tau_B$, integrating this dissipation yields the total history energy, precisely the Consensus Action $\mathcal{S}_{\text{PoW}}$ [Eq. (31)]. To ground this energy in physical mass, we normalize it by the hardware's embodied specific exergy e_{mfg} [$\text{J}\,\text{kg}^{-1}$] [Sec. III B]:

$$M_{\text{eff}}^{(\text{phen})} \equiv \frac{1}{e_{\text{mfg}}\,\tau_B} \int_{t_1}^{t_2} \bar{E}_{\text{eff}}(t)\,dt = \frac{\mathcal{S}_{\text{PoW}}(t_1,t_2)}{e_{\text{mfg}}\,\tau_B}\,. \quad (63)$$

We must introduce a strict formal distinction between the **Global Mass** of history and the **Marginal Mass** of consensus. Equation (63) defines a phenomenological global inertia scale, $M_{\text{global}}^{(\text{phen})} \equiv M_{\text{eff}}^{(\text{phen})}$, corresponding to the integrated volume of the potential well (the ledger's total thermodynamic inertia since the genesis block).

However, the network's resistance to a *local* perturbation—such as a reorganization attempt at the active chain tip—is not dictated by this historical integral, but rather by the local curvature of the Landau potential at the minimum of the well. By analogy with massive field theories, we isolate this restoring force at the phase boundary by defining an energetic curvature parameter κ_{marginal}, associated with the second derivative of the potential well evaluated at the equilibrium order parameter Φ_0:

$$\kappa_{\text{marginal}} \equiv \frac{1}{2}\frac{\partial^2 \mathcal{U}}{\partial \Phi^2}\bigg|_{\Phi_0} = -2\,\alpha_{\text{h}}(T_{\text{eff}})\,. \quad (64)$$

Because $\alpha_{\text{h}}(T_{\text{eff}}) < 0$ in the ordered phase, κ_{marginal} is strictly positive and carries the dimensions of energy ($[\text{M}\,\text{L}^2\,\text{T}^{-2}]$).

To phenomenologically relate this energetic stiffness to the topological mass gap $m = \xi^{-1}$ defined in Sec. IV B, we bridge the system's macro-canonical probabilities to its energy landscape. In an effectively Boltzmann-distributed regime, the relative statistical weight of the

canonical history against a competing fork (p/q) scales with the system's energetic restorative force normalized by thermal noise. Mapping the hashrate ratio to this localized energetic stiffness yields

$$\xi^{-1} \sim \ln\left(1 + \frac{\kappa_{\text{marginal}}}{k_B T_{\text{eff}}}\right). \tag{65}$$

Because the ratio $\kappa_{\text{marginal}}/(k_B T_{\text{eff}})$ is purely dimensionless, the resulting expression preserves dimensional consistency and ensures the topological correlation length ξ remains a dimensionless block count.

We note that this identification is valid in the near-critical regime $p/q \approx 1$, where $\ln(p/q) \approx (p-q)/q$. Deep in the ordered phase ($p \gg q$), logarithmic corrections become significant and the linear relation breaks down.

Equating this to our earlier probabilistic derivation of the mass gap ($m = \ln(p/q)$) reveals a suggestive thermodynamic symmetry. Through this phenomenological mapping, we obtain the following relation for network security:

$$\frac{p-q}{q} = \frac{\kappa_{\text{marginal}}}{k_B T_{\text{eff}}}. \tag{66}$$

This indicates that the relative hashrate advantage of the honest network functions as an effective **thermodynamic signal-to-noise ratio**—the energetic restoring force localized at the chain tip divided by the active thermal noise. It is this energetic marginal mass, dynamically tuned by the protocol's difficulty target, that dictates the macroscopic correlation length ξ. Thus, while Global Mass anchors the blockchain in irreversible thermodynamic history, Marginal Mass protects the active chain tip against thermal evaporation. As established in Sec. IV, this mass gap acts as an emergent dynamical inertia, functioning exclusively as an **effective thermodynamic resistance to temporal reversibility**.

In light of Landauer's Principle [4], Proof of Work is not an arbitrary mathematical puzzle; it is a massive, continuous process of **uncertainty erasure**. Every second, the network explores, evaluates, and physically destroys hundreds of exahashes of invalid microstates. The blockchain's colossal phenomenological inertia scale $M_{\text{eff}}^{(\text{phen})}$ may therefore be interpreted as the macroscopic remanence of the astronomical number of erased cryptographic failures required to distill a singular canonical history.

It is precisely the irreversibility of this relentless informational erasure that endows the ledger with its inertia. Rather than curving spacetime, this uninterrupted accumulation of dissipated heat sculpts a potential well in the free-energy landscape $\mathcal{F}$ so remarkably deep that the thermodynamic "escape velocity" required to breach the $\Delta\mathcal{F}$ barrier and rewrite history macroscopically exceeds the total energetic capacity of any individual actor. Ultimately, Nakamoto consensus operates not merely as a cryptographic software agreement, but as a physical attractor characterized by formidable thermodynamic inertia.

C. Asymptotic Physical Limits and Broad Applicability

As a rough consistency check, one may estimate the asymptotic floor of h_N by combining the Landauer limit at $T_{\text{sub}} = 300\,\text{K}$ ($k_B T_{\text{sub}} \ln 2 \approx 2.87 \times 10^{-21}\,\text{J}$ per irreversible bit operation, with $\sim 10^5$ such operations per SHA-256 hash) with femtosecond-scale switching times, yielding $h_{N,\text{limit}} \sim 10^{-31}\,\text{J}\cdot\text{s}$—within a few orders of magnitude of Planck's constant h. However, material constraints intervene well before this limit: at the current computational mass efficiency $\mu_m \approx 1.5 \times 10^{-13}\,\text{kg}\cdot\text{s}$, exploiting the full solar irradiance ($\sim 10^{17}\,\text{W}$) would demand a hardware mass exceeding $10^{19}\,\text{kg}$, indicating that material, rather than purely energetic, bounds will dominate in practice.

The protocol may thus be interpreted as anchoring cryptographic security in physical resource constraints and irreversible dissipation. By formalizing consensus as a non-equilibrium phase transition, this framework yields falsifiable limits, including an absolute bound on block volume ($V_{\text{B,crit}}$) to avert topological collapse and predicted variance divergences during exergetic quenches.

It would be interesting to explore whether analogous dissipative ordering mechanisms appear in other systems sustained far from equilibrium, such as neural synchronization or cellular proofreading, though such extensions lie beyond the scope of the present work. Notably, this framework mathematically parallels recent observations linking the thermodynamics of Nakamoto consensus to cellular biology, where ribosomes utilize kinetic proofreading as a biological "proof-of-work" to reduce thermal noise and assemble proteins accurately from RNA code [32].

More modestly, the present formalization places the distributed ledger at an instructive intersection of thermodynamics, information theory, and decentralized computation. By coupling cryptographic asymmetry [33] with Shannon information [8] and irreversible dissipation, the framework suggests how large-scale consensus can be analyzed as an active ordering process subject to energetic, informational, and material constraints. In that sense, Nakamoto consensus may serve as a useful laboratory for studying how decentralized architectures resist entropic degradation under sustained non-equilibrium forcing.

ACKNOWLEDGMENTS

The authors wish to thank the open-source community, the founding cypherpunks, and researchers exploring the intersection of physics and decentralized systems. The code used to generate the empirical plots and the raw data presented in Table II are available in a permanent repository at: `https://github.com/pascalranaora/NakamotoRanaoraPRE2026`.

Appendix A: Summary of Physical Variables. Principal variables and dimensional assignments used in the phenomenological model. Assignments are chosen to maintain consistency across the 1D Landau theory, global-mode stochastic dynamics, and action-based quantities.

Sym.	Description	Sec.	Dimension	SI
	Temporal and kinematic variables			
t	External physical time	I	Time $[T]$	s
$\mathbf{B}$	Chain configuration	II.A	Set [1]	–
B_i	Individual block state	II.A	Structure [1]	–
N	Ledger time (block height)	II.A	Count [1]	blocks
τ_B	Target block interval	II.C	Time $[T]$	s
τ_L	Network propagation latency	II.C	Time $[T]$	s
τ_{hop}	Single-hop latency	III.A	Time $[T]$	s
τ_0	Geometric delay floor	III.A	Time $[T]$	s
$H(t)$	Global network hashrate	III.D	$[T^{-1}]$	s^{-1}
τ_{hash}	SHA-256 transition time	III.E	Time $[T]$	s
z	Reorganization depth	IV.A	Count [1]	blocks
r_{tech}	Innovation rate	VII.A	$[T^{-1}]$	s^{-1}
η	Exergetic growth exponent	VII.A	[1]	–
	Thermodynamic and energetic variables			
$E_{\text{eff}}(B_i)$	Stochastic block energy	II.A	Energy	J
$\Sigma(\mathbf{B})$	Dissipation functional	II.A	Energy	J
β_{eff}	Inverse eff. thermal energy	II.A	Energy^{-1}	J^{-1}
T_{sub}	Substrate temperature	II.B	Temperature	K
T_{bath}	Exergy-bath temperature	VII.B	Temperature	K
$\bar{E}_{\text{eff}}$	Mean energy per block	II.C	Energy	J
k_B	Boltzmann constant	II.C	E/T	J K^{-1}
T_{eff}	Eff. information temperature	II.C	Temperature	K
$\mathcal{F}$	GL free energy	II.C	Energy	J
S_{th}	Thermodynamic entropy	II.C	E/T	J K^{-1}
$\Phi(t)$	Global order parameter	II.D	[1]	–
T_c	Critical eff. temperature	II.D	Temperature	K
$\dot{Q}_{\text{diss}}$	Heat dissipation rate	II.E	Power	W
$\phi(x,t)$	Local order parameter	III.B	[1]	–
$f(\phi)$	Local free-energy density	III.B	E/L	J m^{-1}
κ	Gradient stiffness (1D)	III.B	$\text{E}\cdot\text{L}$	J m
a, b	Landau coefficients	III.B	E/L	J m^{-1}
c_0	Saturation structural const.	III.B	[1]	–
χ_E	Thermal susceptibility	III.B	$\text{E}/(\text{L}\cdot\text{T})$	$\text{J m}^{-1}\text{K}^{-1}$
$\mathcal{E}_{\text{coupling}}$	Coupling energy density	III.B	E/L	J m^{-1}
ϕ_0	Equilibrium hashrate state	III.B	[1]	–
e_{mfg}	Embodied ASIC exergy	III.B	E/M	J kg^{-1}
$e_{\text{hash}}(t)$	Energy per hash	III.D	Energy	J
$P_{\text{net}}(t)$	Total network power	III.D	Power	W
σ_{obs}	Entropy production (window)	III.D	E/T	J K^{-1}
$\Delta\mathcal{F}(z)$	Reorg. free-energy barrier	IV.C	Energy	J
$\bar{J}$	XY-model coupling	VI.A	Energy	J
E_{vortex}	Vortex energy	VI.C	Energy	J
S_{vortex}	Vortex config. entropy	VI.C	E/T	J K^{-1}
T_{KT}	KT transition temperature	VI.C	Temperature	K
	Thermodynamic and energetic variables (cont.)			
$\mu(t)$	Exergetic chemical potential	VII.A	Energy	J
μ_{exergy}	Exergy component of μ	VII.A	Energy	J
$\mathcal{U}(\Phi)$	Homogeneous eff. potential	VII.B	Energy	J
$\alpha_{\text{h}}, b_{\text{h}}$	Homogeneous Landau coeff.	VII.B	Energy	J
Λ_0	Relaxation mobility	VII.B	$\text{E}^{-1}\text{T}^{-1}$	$\text{J}^{-1}\text{s}^{-1}$
$\mathcal{A}(t)$	Anticipatory driving term	VII.B	$[T^{-1}]$	s^{-1}
g_{ant}	Anticipation coupling	VII.B	$\text{E}^{-1}\text{T}^{-1}$	$\text{J}^{-1}\text{s}^{-1}$
τ_{plan}	Planning horizon	VII.B	Time $[T]$	s
λ	Temporal discount rate	VII.B	$[T^{-1}]$	s^{-1}
H_{eq}	Adiabatic equil. hashrate	VII.B	$[T^{-1}]$	s^{-1}
X	Fluctuation–dissipation ratio	VII.B	[1]	–
c_m	Volatility scaling constant	VII.B	[1]	–
T_{act}	Eff. activity temperature	VII.B	Temperature	K
$\zeta(t)$	Active noise term	VII.B	$[T^{-1}]$	s^{-1}
D_{act}	Active diffusion scale	VII.B	$[T^{-1}]$	s^{-1}
$\dot{S}_i$	Entropy production rate	VII.B	$\text{E}/(\text{T}\cdot\text{T})$	$\text{J K}^{-1}\text{s}^{-1}$
Δt	Inter-block wait time	VII.C	Time $[T]$	s
	Information and topological variables			
Ω	State-space of forks	II.A	Count [1]	–
$\vec{s}_i$	Local consensus spin	II.D	[1]	–
Γ, Γ_c	Latency ratio / critical val.	III.A	[1]	–
$V_B, V_{\text{B,crit}}$	Block size / critical size	III.A	Information	Bytes
C_{node}	Bottleneck link capacity	III.A	Info/T	B s^{-1}
γ	Validation-time coeff.	III.A	T/Info	s B^{-1}
D_{hop}	Network diameter (hops)	III.A	Count [1]	hops
C_{eff}	Eff. global capacity	III.A	Info/T	B s^{-1}
L_{net}	Network spatial diameter	III.B	Length $[L]$	m
$\mathcal{D}$	Protocol difficulty	III.D	[1]	–
$\mathcal{N}_H(\mathcal{D})$	Expected hashes per block	III.D	Count [1]	hashes
$G(z)$	Two-point correlation fn.	IV.A	[1]	–
$\xi, \xi_{\parallel}$	Correlation length	IV.A	Topo. dist. [1]	blocks
p, q	Honest / attacker share	IV.B	Proportion [1]	–
$P(z)$	Fork survival probability	IV.D	Probability [1]	–
$\mathcal{V}, \partial\mathcal{V}$	Volume and Markov blanket	V.A	State set [1]	–
S_{Shannon}	Shannon entropy	V.B	Information	bits
θ_i	Consensus phase variable	VI.A	Phase [1]	rad
$\mathcal{N}_{\text{nodes}}(r)$	Nodes within radius r	VI.C	Count [1]	–
L, a_0	Diameter / lattice const.	VI.C	Topo. dist. [1]	hops
d_{eff}, d_s	Effective / spectral dim.	VI.C	[1]	–
	Action and inertia variables			
μ_m	Computational mass efficiency	III.B	$\text{M}\cdot\text{T}$	kg s
$h_N(t)$	Nakamoto action per hash	III.E	Action	J s
$\mathcal{S}_{\text{PoW}}$	Accumulated consensus action	III.E	Action	J s
$m{=}\xi^{-1}$	Inverse correlation depth	IV.B	Inv. topo. dist.	blocks^{-1}
$M_{\text{eff}}^{(\text{phen})}$	Phenomenological inertia	VIII.B	Mass	kg
κ_{marginal}	Potential curvature scale	VIII.B	Energy	J
M_{hardware}	Physical hardware mass	VIII.C	Mass	kg

Appendix B: Plan of the Article & Logical Flow. Compact summary of each section and subsection, indicating how the phenomenological framework progresses from foundational concepts to coarse-grained dynamics and empirical observations.

ID	Section / Subsection Name	Physical Summary / Theoretical Role
	I. Introduction: The Thermodynamics of Distributed Consensus	
Introduces consensus as an effective symmetry-breaking process sustained by dissipation rather than purely logical ordering.		
I.	Scope, Assumptions, and Limitations	Clarifies the phenomenological status of the model and distinguishes operational claims from heuristic analogies.
	II. Network Model, Statistics and the Arrow of Time	
Maps ledger evolution to a driven non-equilibrium system weighted by cumulative dissipation.		
II.A	The Minimal Network Model and Consensus Hamiltonian	Defines the blockchain as a 1D growing network and introduces the dissipation-weighted path measure.
II.B	Time-Reversal Symmetry and Sybil Degeneracy	Interprets costless identity proliferation as a high-degeneracy, high-temperature regime.
II.C	Information Temperature and Thermodynamic Regimes	Defines T_{eff} from the ratio of latency to block interval and identifies ordered/disordered regimes.
II.D	Spontaneous Symmetry Breaking (SSB)	Introduces the global order parameter and the onset of macroscopic alignment.
II.E	Emergence of Thermal Time and Fluctuation Theorem	Uses heuristic thermal-time scaling and fluctuation relations to motivate irreversible ledger time.
	III. Thermodynamic Potential and Symmetry Breaking	
Constructs a coarse-grained Landau description of consensus stability and protocol-dependent limits.		
III.A	Effective Temperature, Microscopic Latency, and the Block Size Limit	Derives an effective thermal bound on block size from propagation, bandwidth, and validation friction.
III.B	The Order Parameter and Landau Free Energy	Introduces the 1D Ginzburg–Landau functional and identifies how latency controls ordering.
III.C	Phase-like Fluctuations and Consensus Rigidity	Describes local branch disagreements as phase-like fluctuations penalized by spatial stiffness.
III.D	Macroscopic Energy Scale, the DAA Thermostat	Interprets the DAA as a feedback mechanism stabilizing the effective energy scale.
III.E	The Nakamoto Action and Hardware Substrate	Defines the action-based hardware anchor linking microscopic switching events to macroscopic consensus.
	IV. Correlation Length and Probabilistic Finality	
Describes finality as an exponential decay of topological correlations with depth.		
IV.A	The Correlation Function	Defines finality through tip-to-history correlations.
IV.B	Derivation of the Mass Gap (The "6-Block" Rule)	Relates reorganization suppression to an effective inverse correlation depth.
IV.C	Nucleation of a False History	Interprets history rewriting as activation over a depth-dependent free-energy barrier.
IV.D	Forks as Absorbing State Transitions (Directed Percolation)	Treats orphaned forks as active states decaying toward an absorbing canonical history.
	V. Information Entropy and Markov Blanket	
Explains how the protocol compresses history into a reduced state description under finite communication constraints.		
V.A	The Volume, the Markov Blanket, and RG Flow	Describes validation as an irreversible coarse-graining onto the active UTXO state.
V.B	Shannon vs. Thermodynamic Entropy	Separates informational uncertainty from dissipative cost at hardware and consensus scales.
V.C	The UTXO Set as a Physical Maxwell's Demon	Uses the demon analogy to interpret filtering and erasure costs phenomenologically.
V.D	Blanket Saturation and Channel Limit	Identifies synchronization loss when update volume approaches effective channel capacity.
	VI. Topological Stability of the Network Graph	
Studies transverse consensus stability on the peer-to-peer graph using phase ordering and topological defects.		
VI.A	The Consensus Phase Field	Assigns node phases and coupling energies to represent spatial synchronization.
VI.B	Forks as Topological Defects (Vortices)	Models persistent partitions as topological defects in the effective phase field.
VI.C	Effective Dimensionality and the Kosterlitz–Thouless Bound	Uses a reduced-dimensional limit to describe partition-driven loss of coherence.
	VII. Non-Equilibrium Dynamics: Exergetic Expansion and the "Halving" Quench	
Analyzes temporal response to deterministic reward shocks in a driven non-equilibrium setting.		
VII.A	The Exergy-Driven Potential	Models long-run hashrate growth through an exergy-limited secular trend.
VII.B	Non-Equilibrium Stochastic Dynamics and FDT Violation	Introduces a global-order-parameter Langevin model with anticipatory forcing and active noise.
VII.C	Thermodynamic Halving Quench and Thermostat (DAA)	Uses sensitivity analysis of Halving windows to characterize non-equilibrium response and damping.
VII.D	Testable Predictions and Operational Diagnostics	Summarizes the principal falsifiable consequences of the framework.
	VIII. Conclusion: The Physics of Distributed Consensus	
Summarizes the framework, its limits, and its broader interpretive scope.		
VIII.A	Model Summary	Recaps the main thermodynamic and information-physics perspectives.
VIII.B	Thermodynamic Inertia and a Phenomenological Mass Analogy	Introduces a cautious mass-like analogy for resistance to history reversal.
VIII.C	Asymptotic Physical Limits and Broad Applicability	Discusses energetic and material limits and possible extensions to other dissipative systems.

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